

3. Precise typing of pattern matching

Accepted type of a pattern p : $\llbracket p \rrbracket = \{v \mid v \text{ matches } p\}$, e.g., $\llbracket (x, y) \rrbracket = \mathbb{1} \times \mathbb{1}$

Boolean type connectives are needed to type pattern matching:

$\text{match } e \text{ with } p_1 \rightarrow e_1 \mid p_2 \rightarrow e_2 \text{ with } e : T$

- To infer the type T_1 of e_1 we need $T \wedge \llbracket p_1 \rrbracket$;
- To infer the type T_2 of e_2 we need $(T \setminus \llbracket p_1 \rrbracket) \wedge \llbracket p_2 \rrbracket$;
- The type of the match expression is $T_1 \vee T_2$.
- Pattern matching is exhaustive if $T \leq \llbracket p_1 \rrbracket \vee \llbracket p_2 \rrbracket$.

Formally:

$$\frac{\begin{array}{c} \text{[Match]} \\ \Gamma \vdash e : T \quad \Gamma, T \wedge \llbracket p_1 \rrbracket / p_1 \vdash e_1 : T_1 \quad \Gamma, T \setminus \llbracket p_1 \rrbracket / p_2 \vdash e_2 : T_2 \end{array}}{\Gamma \vdash \text{match } e \text{ with } p_1 \rightarrow e_1 \mid p_2 \rightarrow e_2 : T_1 \vee T_2} \quad (T \leq \llbracket p_1 \rrbracket \vee \llbracket p_2 \rrbracket)$$

where T/p is the type for the capture variables in p when it is matched against values in T .